

e) The solution set of $Ax=b$ is the set of all vectors of the form $w=p+V_h$, where V_h is any solution of the equation $Ax=0$.

Solution: On page 47, Theorem 6 is mentioned as follows: Suppose the equation $Ax=b$ is consistent for some given b , and let p be a solution. Then the solution set of $Ax=b$ is the set of all vectors of the form $w=p+V_h$, where V_h is any solution of the homogeneous equation $Ax=0$.

The given statement is incomplete, hence False $\square$

24a) If x is a nontrivial solution of $Ax=0$, then every entry in x is non-zero.

Solution: On page 44, just above the topmost grey box, it is mentioned that a nontrivial solution is a nonzero vector x that satisfies $Ax=0$. A non-zero vector means at least one entry is non-zero.

$\therefore$ The given statement is False $\square$

b) The equation $x=x_2u+x_3v$, with x_2 and x_3 free (and neither u nor v a multiple of the other), describes a plane through the origin.

Solution: By ~~using~~ the arguments presented in Example 2 (Pg-44 and 45), we can see that the given statement is True.

c) The equation $Ax=b$ is homogeneous if the zero vector is a solution.

Solution: From page 43 under the section "Homogeneous Linear Systems" it is mentioned that a system of linear equations is said to be homogeneous if it can be written in the form $Ax=0$, where A is an $m \times n$ matrix and 0 is the zero vector in $\mathbb{R}^m$.

$\therefore$ The given statement is False $\square$

d) The effect of adding p to a vector is to move the vector in a direction parallel to p .

Solution: On page 46 just below ~~the~~ Example 3, it is mentioned that